



Intelligent University Course Finder



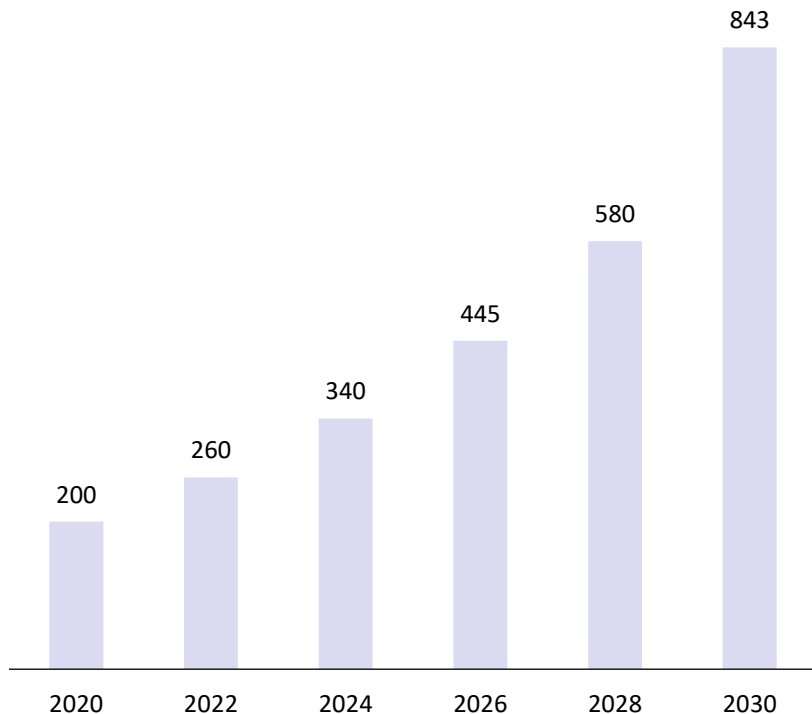
FDE2ndBatchTraining Capstone

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<https://github.com/YuichiYamazaki-source/Capstone>

MARKET OPPORTUNITY

E-Learning Market is Booming



Why It's Growing

- **AI Integration**

60% of teachers use AI; adaptive learning platforms growing 41% YoY

- **Remote Work Shift**

Corporate e-learning demand surging as 60% of workforce needs reskilling

- **Mobile Access**

Smartphone-first learning in Asia-Pacific driving fastest regional growth

- **Cost Advantage**

Online courses up to 50-70% cheaper than traditional education



The Problem

Universities offer a wide range of courses, but students struggle to find courses that truly match their interests, background, and career goals. Traditional keyword search fails to capture real intent.



Intent Mismatch

Students describe learning interests in natural language — not course names or catalog codes.



Prerequisite Confusion

Understanding prerequisite relationships and identifying knowledge gaps before advanced courses.



Path Discovery

Discovering structured learning paths and career-aligned course sequences is nearly impossible.



Fragmented Info

Course info is spread across multiple formats — catalogs, advising guides, and department docs.

Project Positioning — PoC Complete + PoB Partial

This project successfully demonstrates **technology feasibility** (PoC), but stops short of full Proof of Business — and for good reason. Three structural barriers make a complete PoB premature at this stage.



Data Quality Wall

Course descriptions are only a few sentences long — too short for precise, reliable skill-to-course matching.



Ground Truth Problem

38 test cases are DB-query-derived, not human-validated. Even the developer cannot confidently label the "right" course.



No Real User Data

Solo development means a sample size of one. Personalization cannot be validated without real users in the system.



Why This Framing Matters

Acknowledging these gaps is a sign of rigor — not weakness. Claiming full PoB prematurely would undermine credibility when the system is stress-tested against real-world data.

Demo

Architecture

Explain by [Architecture.pdf](#)

System Requirements

REQUIREMENT 1

Basic Tier

- Basic RAG for course discovery
- Intent-based semantic search
- Simple recommendation agent
- Skill gap identification
- Difficulty level filtering
- Learning objective validation guardrails
- Basic course sequencing
- Metadata filtering (org, rating, difficulty)



Expose via API endpoint

REQUIREMENT 2

Advanced Tier

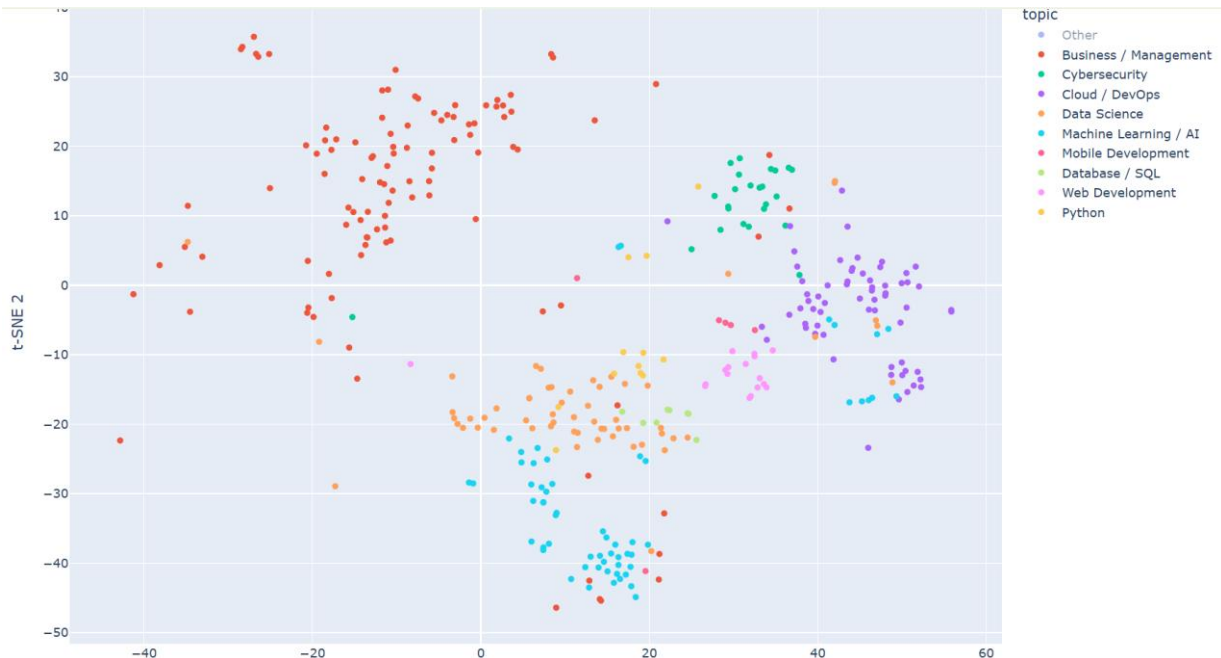
- DeepEval for recommendation relevance and learning outcomes
- Rerank using learner preference models and success rate data
- LLM-as-judge for course quality and prerequisite validation
- Token optimization for personalized learning path generation
- Performance testing: real-time recommendations at scale
- Content appropriateness and prerequisite guardrails



**Simple front-end UI
for end-user demonstration**

EMBEDDING ANALYSIS

Topic → Clear Clusters in Embedding Space



Embedding input:

title + description only

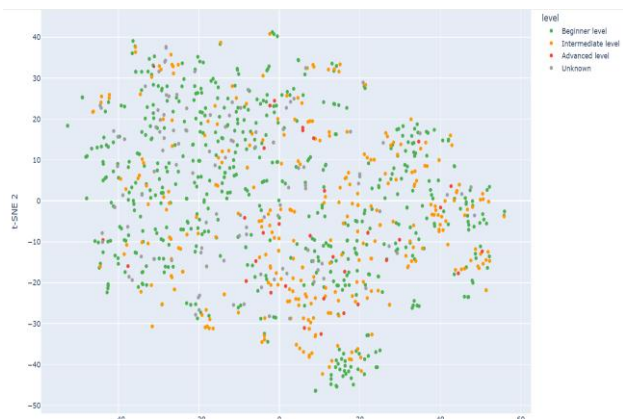
Result:

Topic/subject clearly captured by semantic similarity.

Business vs. Tech domains separate cleanly.

EMBEDDING ANALYSIS

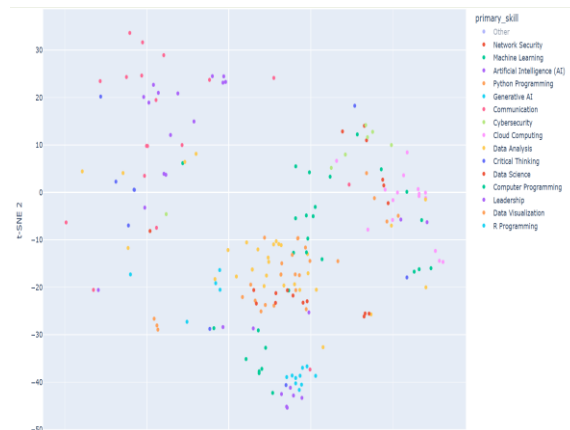
No Clusters



Level



Organization



Skill

Why?

Embedding uses title + description only. Level, organization, and granular skills are **not encoded** in the embedding space.

→ These dimensions require payload filters and keyword (BM25) matching.

DESIGN DECISION

Data Analysis → Hybrid Search Design

Dimension	Captured?	Evidence	RAG Handling
Topic / Subject	✓ Yes	Clear t-SNE clusters	Dense vector (semantic search)
Specific Skills	△ Partial	Sparse data (29.4% missing)	BM25 keyword matching
Difficulty Level	✗ No	Uniform distribution	Qdrant payload filter
Organization	✗ No	Scattered across space	Qdrant payload filter

Conclusion: Embedding captures WHAT the course is about. Metadata (level, org, rating) needs structural filtering.

Strong Points of This Capstone

Three pillars that demonstrate engineering rigor, observability, and disciplined scope management throughout the project lifecycle.

1

End-to-End Working PoC

Hybrid Search (BM25 + Dense + RRF) paired with a Multi-Agent system (4 LLMs, 5 Tools) and a React frontend — all wired into a single pipeline: user query → agent routing → hybrid retrieval → LLM response. One-command startup via Docker Compose.

2

Measurement-First Design

LangFuse tracing instruments every LLM call (model, tokens, cost, latency). DeepEval covers Answer Relevancy, Faithfulness, and Actionability. IR metrics include Hit Rate, Precision@K, and Recall@K — all built *before* feature expansion.

3

Scope Discipline

PoC / PoB / Prd boundaries explicitly defined. "What we didn't do and why" is fully documented — 7 deferred items, each with a recommended phase and clear rationale.

Challenges & Lessons Learned

What we discovered building the PoC — and what it means for how we move forward.

Biggest Challenge: Ground Truth

- 38 test cases generated from DB queries — not human-validated
- Course descriptions are too short to confidently label "correct" answers
- This remains the **most critical open problem** to solve

Key Lesson

Build observability and evaluation infrastructure **before** expanding features. If you can't measure it, you can't improve it.

Evaluation frameworks must be established early — not retrofitted after the fact.

By the Numbers

A snapshot of the technical depth packed into this capstone.

4

LLMs Integrated

Coordinated via multi-agent routing

5

Agent Tools

Specialized capabilities per tool

3

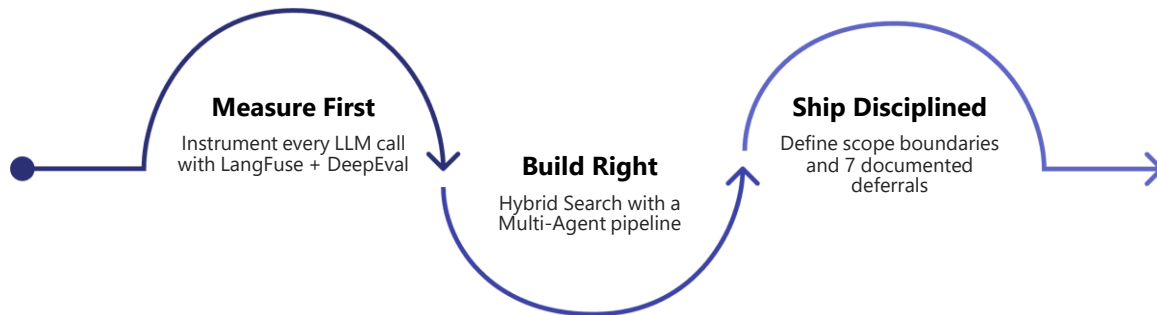
Retrieval Metrics

Hit Rate, Precision@K, Recall@K

7

Deferred Items

Each with phase & justification



The capstone follows a deliberate sequence: establish measurement infrastructure, build the full pipeline, then enforce scope boundaries — a pattern that mirrors production-grade engineering practice.

Next Action

Business Strategy & Validation

Technology alone is not enough — define how to **measure and prove business value**.



ROI Calculation

API + infra cost model vs. value delivered
— time saved, course match quality



Market Analysis

Competitive landscape + TAM/SAM/SOM for AI-powered course discovery



PoC vs. PoB Criteria

PoC = tech feasibility. PoB = business value.
Clear go/no-go thresholds.



KPI Definition

Engagement rate, recommendation acceptance, skill gap closure



Production-Ready

means hitting defined performance thresholds, accuracy targets, and cost ceilings — not just deploying to the cloud.

Agent & AI Optimization

Goal: Optimize cost, accuracy, latency, and capability through data-driven iteration.



Caching & Compression

- Cache repeated query patterns
- Prompt compression via context pruning
- Reduce latency and API token cost



Model Routing

- Cost Management
- Insurance Quality
- Remain Userbility



Agent Refinement

- Optimize handoff logic
- Error handling & fallback strategies
- Multimodal: image/PDF input support



ML Integration

- Recommendation engine
- User behavior analysis
- Personalization layer

Feature Roadmap & Production Infrastructure

Feature Roadmap

Course Analytics Dashboard

Engagement metrics, popularity trends, completion rates

User Profile Enhancement

MCQ-based skill assessment, learning style detection

Course Registration

Admin CRUD interface for managing course catalog

Data Retention & Archival

Structured storage policy, historical data, backups

Secure Cloud Infrastructure

Hosting & CI/CD

Azure App Service (containers) + GitHub Actions Runner

Networking

Integration Runtime — NIC inside Azure VNet, zero public endpoints

Secrets & IaC

Azure Key Vault + Terraform for infrastructure as code

Zero-Trust Security

VNet isolation, Managed Identity, self-hosted agent pool

Additional Considerations for Production

API & Data Quality

API Error Handling

Standardized error codes, retry with backoff, graceful degradation

Input Validation & API Design

Schema validation, rate limiting, versioning, OpenAPI spec

DB Data Quality & Integrity

Missing value handling, data consistency checks, quality metrics

Testing Strategy

Unit / integration / E2E tests + AI response quality evaluation

Governance & Operations

Monitoring & Observability

Log aggregation, APM dashboard, alerting, token usage tracking

Compliance & Privacy

GDPR / data protection, audit logs, consent management

AI Guardrails & Safety

Hallucination detection, content filtering, output validation

Incident Response & Runbooks

On-call playbooks, rollback procedures, SLA definitions

Appendix

Objective

Build an AI-powered multimodal course discovery and recommendation system that acts as an intelligent academic assistant — understanding student intent and delivering personalized, structured guidance.

Multimodal Input

Accepts text, voice, or uploaded documents as query input.

Intent Interpretation

Interprets student learning intent beyond surface-level keywords.

Semantic Retrieval

Retrieves semantically relevant courses from the full catalog.

Intelligent Advisor

Acts as a smart academic assistant guiding every step of the journey.



Key Capabilities

Seven core capabilities power the intelligent course discovery experience — from understanding how a student asks a question to mapping their goals to a career track.



Multimodal Query Understanding

Accepts text, voice, and uploaded document descriptions as input for maximum accessibility.



Semantic Course Retrieval

Goes beyond keyword matching to understand the true meaning behind a student's learning goal.



Relevance Assistance

Explains *why* each course is recommended, building student trust and confidence.



Prerequisite Awareness

Checks whether the student has the required background before recommending advanced courses.



Skill Gap Identification

Suggests foundational courses to fill knowledge gaps before tackling advanced material.

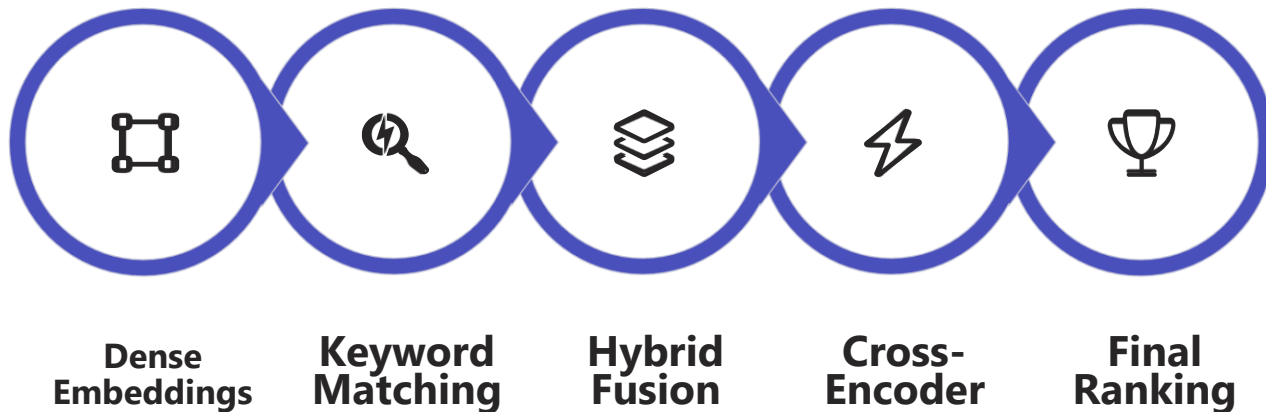


Learning Path Recommendations

Generates structured intro → advanced course sequences tailored to each learner.

Advanced: Hybrid Course Retrieval

The retrieval layer combines the best of vector-based semantic search and traditional keyword matching, then applies cross-encoder reranking to surface the most relevant results.



Vector Embeddings

Dense semantic representations capture meaning, context, and conceptual similarity between queries and course descriptions.

Keyword Retrieval

BM25 lexical matching ensures exact terminology and domain-specific terms are never missed.

Dynamic Filtering

Filter by difficulty, rating, organization, and skill category to narrow results to what matters most.

Cross-Encoder Reranking

A cross-encoder model re-scores candidate courses for improved recommendation quality and precision.

Multi-Agent Architecture

A collaborative multi-agent system where specialized agents handle distinct responsibilities and communicate to deliver a unified, intelligent recommendation experience.

Course Retrieval Agent Retrieves relevant courses from the catalog using hybrid semantic search

Skill Gap Analysis Agent Identifies missing prerequisite skills and recommends foundational courses

Learning Path Planning Agent Generates structured, sequenced course paths from beginner to advanced

Career Alignment Agent Maps recommended courses to specific career tracks (AI, data science, finance, SW)

Learning Advisor Agent Synthesizes all agent outputs into a final, coherent recommendation summary

Additional Intelligence

Analytics & Personalization

Learning Analytics

Tracks popular courses, completion trends, and engagement patterns across the learner population.

Personalized Recommendations

Tailors suggestions based on individual learner preferences, history, and stated goals.

Adaptive Difficulty

Dynamically adjusts recommended difficulty levels based on learner progress and performance signals.

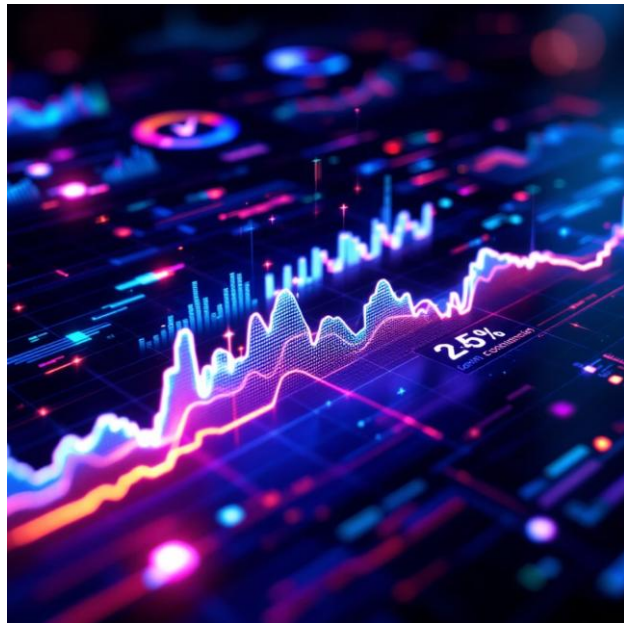
Agent Communication

Agent Handoff

Seamless transfer of context between specialized agents as queries move across domains.

A2A Communication

Agent-to-Agent (A2A) communication enables collaborative filtering and cross-domain reasoning.



Dataset & Deliverables

DATASET

Coursera Course Dataset

Source

Kaggle / HuggingFace /
GitHub

Format

CSV, JSON

Key Fields: course_name, description, skills, difficulty_level, rating, organization

- ❑ Supports semantic retrieval, skill mapping, and learning path generation across domains.

DELIVERABLES

01

Architecture Diagram

Data flow from ingestion to recommendation (JPEG/PDF)

02

Design Document

Embedding model, chunking strategy, agent orchestration decisions and trade-offs

03

Full Executable Code

Modular microservice with README covering setup, ingestion, and example queries

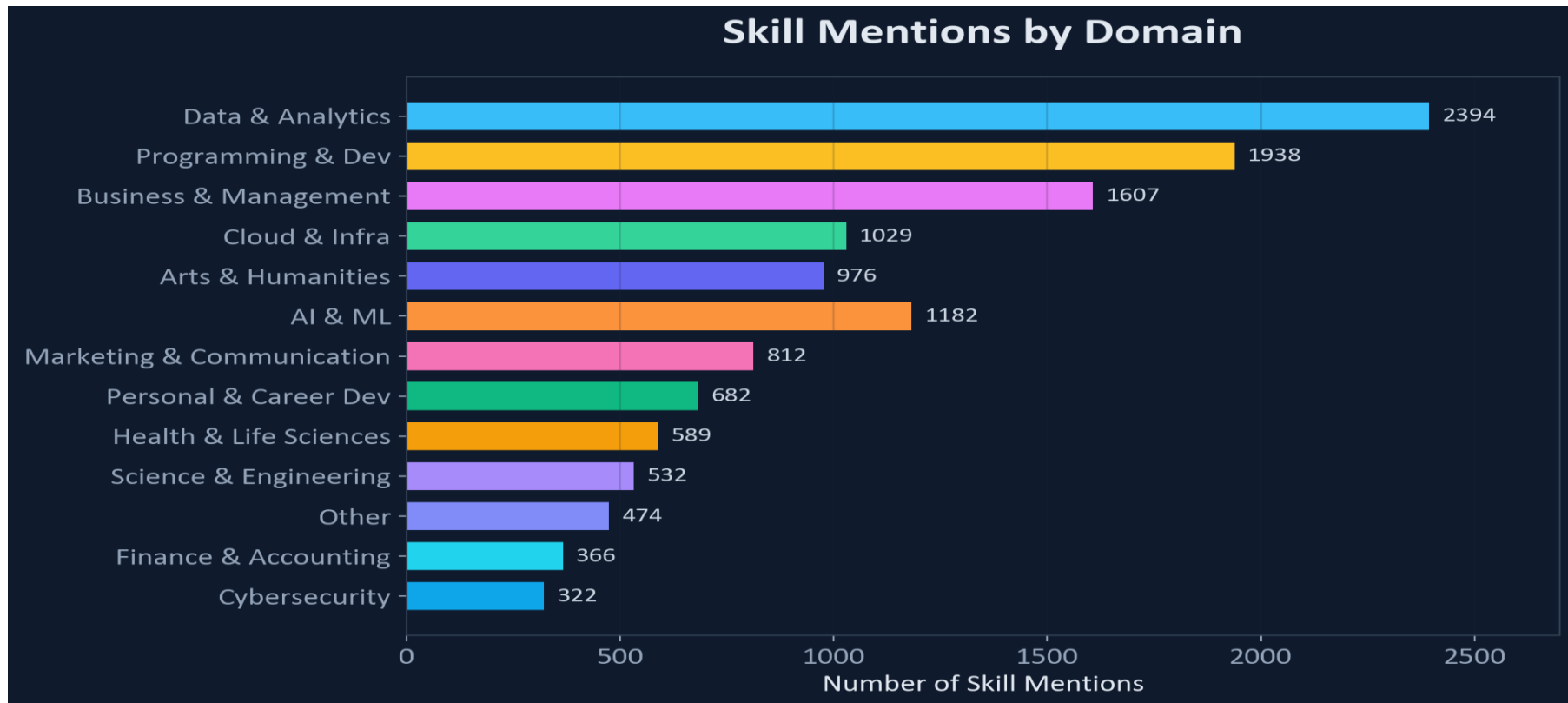
04

Panel Presentation

10 min total — live demo (8 min) + Q&A (2 min)

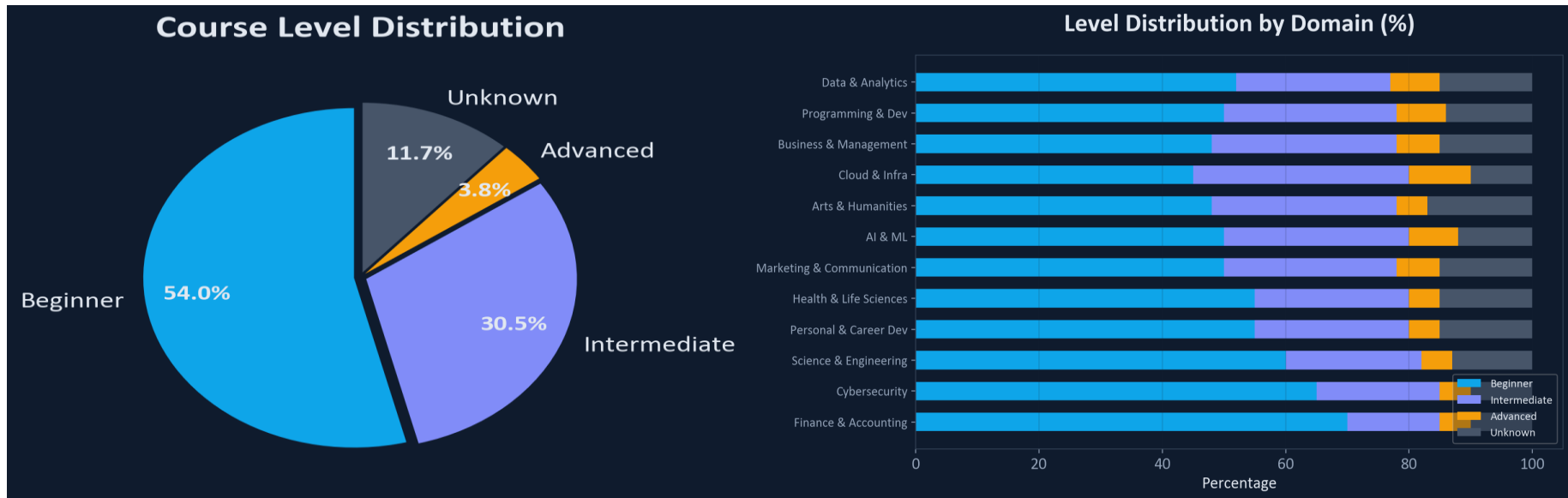
DATA ANALYSIS

6,645 Courses — Domain Distribution



DATA ANALYSIS

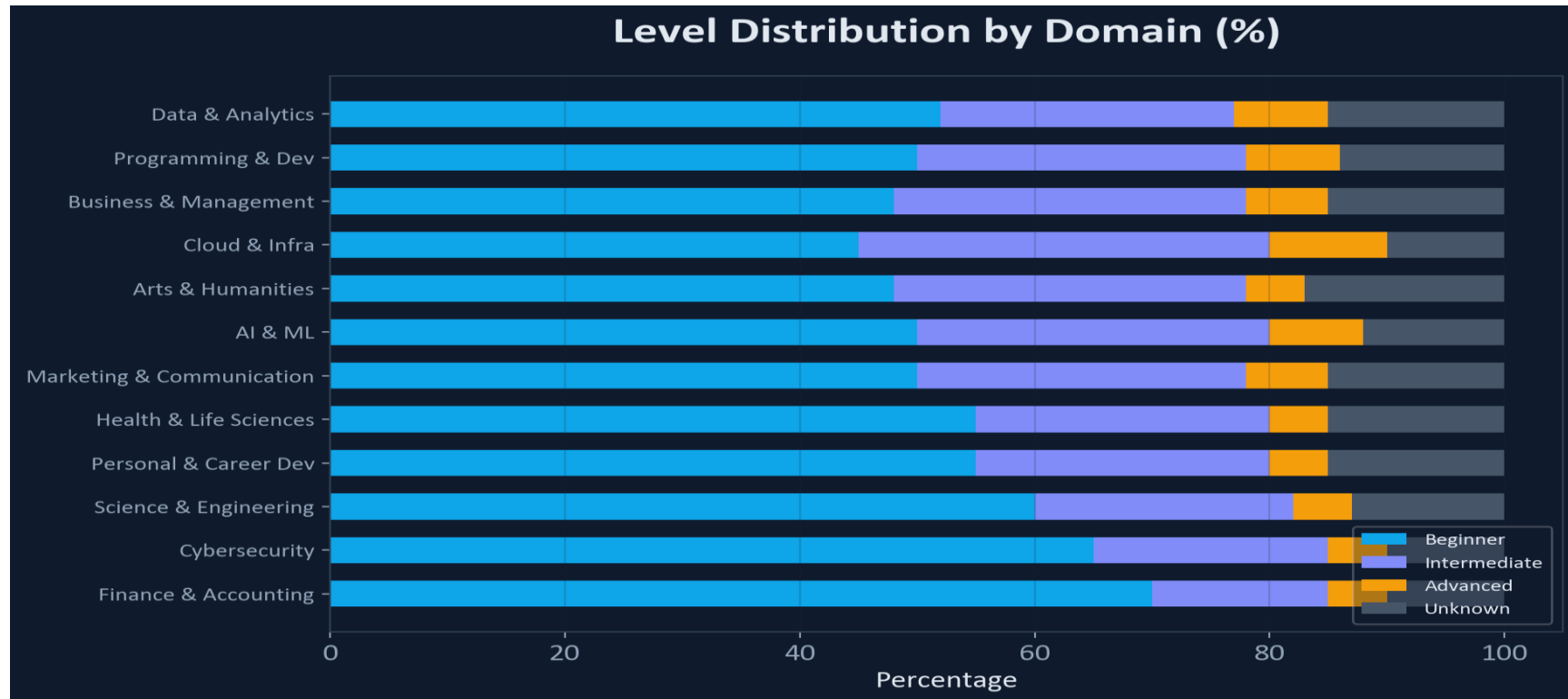
54% Beginner — Consistent Across All Domains



Insight: Beginner-heavy distribution is consistent across domains. Agents must filter by level to surface Advanced content.

APPENDIX

Level Distribution by Domain (Detail)



Indexing Pipeline — Data Ingestion

CSV
6,645 courses

MongoDB
Document store

OpenAI
text-embedding
-3-small

Qdrant
Dense + BM25
vectors

Dense Vector (Semantic)

Input: title + description

Model: text-embedding-3-small

Dimensions: 1,536

BM25 Sparse Vector (Keyword)

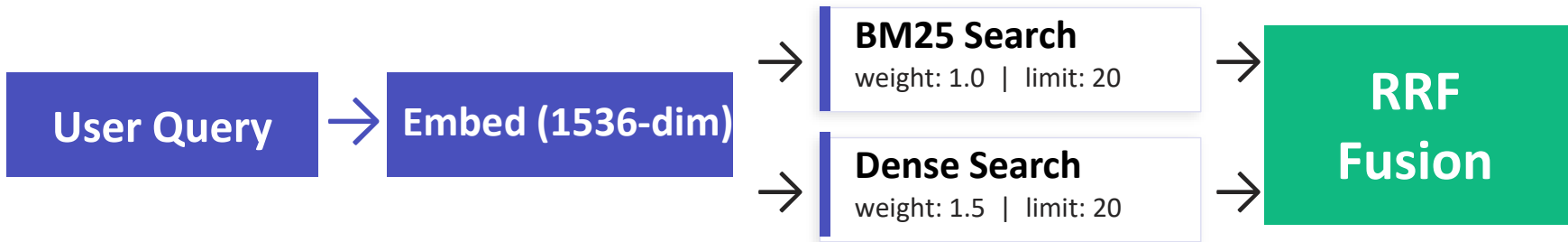
Input: title + description + skills

Model: Qdrant/bm25 (built-in)

Purpose: Exact skill/tool name matching

RAG ARCHITECTURE

Query Pipeline — Hybrid Search + RRF



Payload Filters (applied to both searches)

level (Beginner/Intermediate/Advanced) **min_rating** ($\geq$ value) **organization** (text match)

Why Qdrant?

Native Hybrid Search: BM25 + Dense + RRF in single API call

Payload Filters: Structured metadata filtering on vector results

Docker Native: Lightweight, fits Docker Compose stack

Real-Time Market Data via Web Search



Why Web Search Matters for Agents

Skill Gap Analyst

Searches for current required skills for target roles (e.g., "ML Engineer required skills 2025")

Model: gpt-4o-mini

Career Advisor

Searches for salary data, job market trends, career path information for informed recommendations

Retry: max 2 attempts, exponential backoff

Demo Flow

A five-step end-to-end demonstration showcasing the full intelligence of the system — from a student's first query to a career-aligned course recommendation.



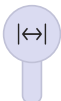
Step 1: Student Query

Student enters a natural language learning goal (e.g., "I want to learn machine learning for finance").



Step 2: Semantic Retrieval

System retrieves the most relevant courses using hybrid vector + keyword search.



Step 3: Skill Gap Analysis

Skill Gap Analysis Agent identifies missing prerequisites and recommends foundational courses.



Step 4: Learning Path Generation

Learning Path Planner generates a structured, sequenced course recommendation from intro to advanced.



Step 5: Career Alignment

Career Alignment Agent maps the final results to relevant career tracks — AI, data science, finance, or software engineering.



Live Demo: 8 minutes of live system interaction followed by 2 minutes of Q&A with the panel.

NEXT STEP

Build the Measurement Framework First

Before proving business value, the foundation for **measuring** that value must be in place. The strategy is to instrument now — so the system is ready to iterate the moment real users arrive.



LangFuse

Online tracing for every LLM call — capturing cost, latency, and behavior in production.



DeepEval

A repeatable evaluation harness enabling consistent, automated quality benchmarking across iterations.



Real Users

With infrastructure in place, each new user adds signal — turning the system into a continuously improving loop.

PoC proves the technology works. The next milestone is clear: onboard real users, collect ground truth, and prove business value.